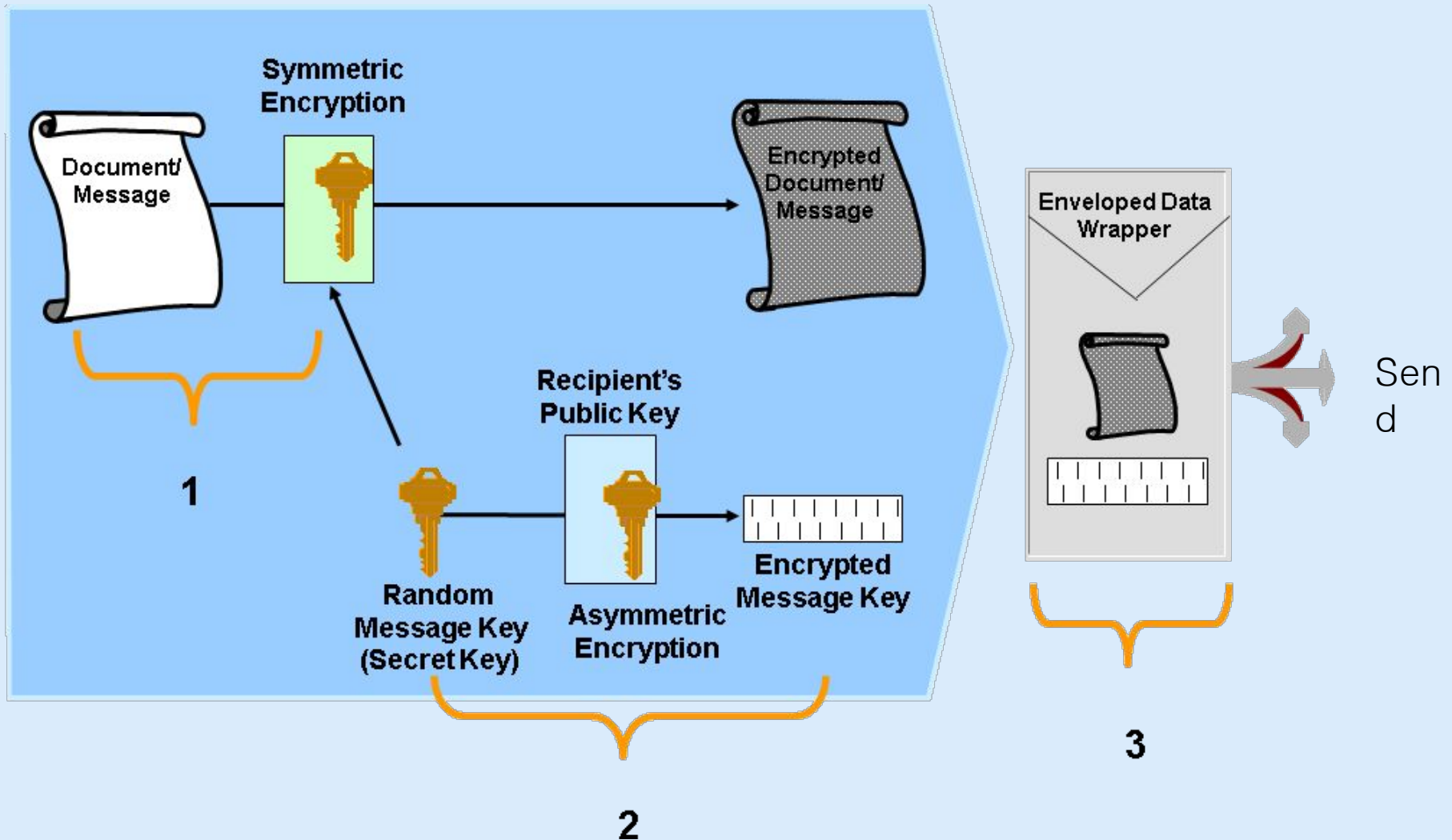


Digital Envelope (Hybrid Encryption)



- PKC is convenient, but much slower to compute compared to symmetric algorithms
 - e.g. in RSA, you have to compute $C^d \bmod n$ where both C and d are very large numbers
- Hybrid encryption allows using the **very efficient symmetric ciphers** without sharing the key in advance
- A random secret key is generated for protecting the message. The key itself is protected using recipient's public key
- Encrypted message and the encrypted secret key are joined together in a wrapper, called digital envelope

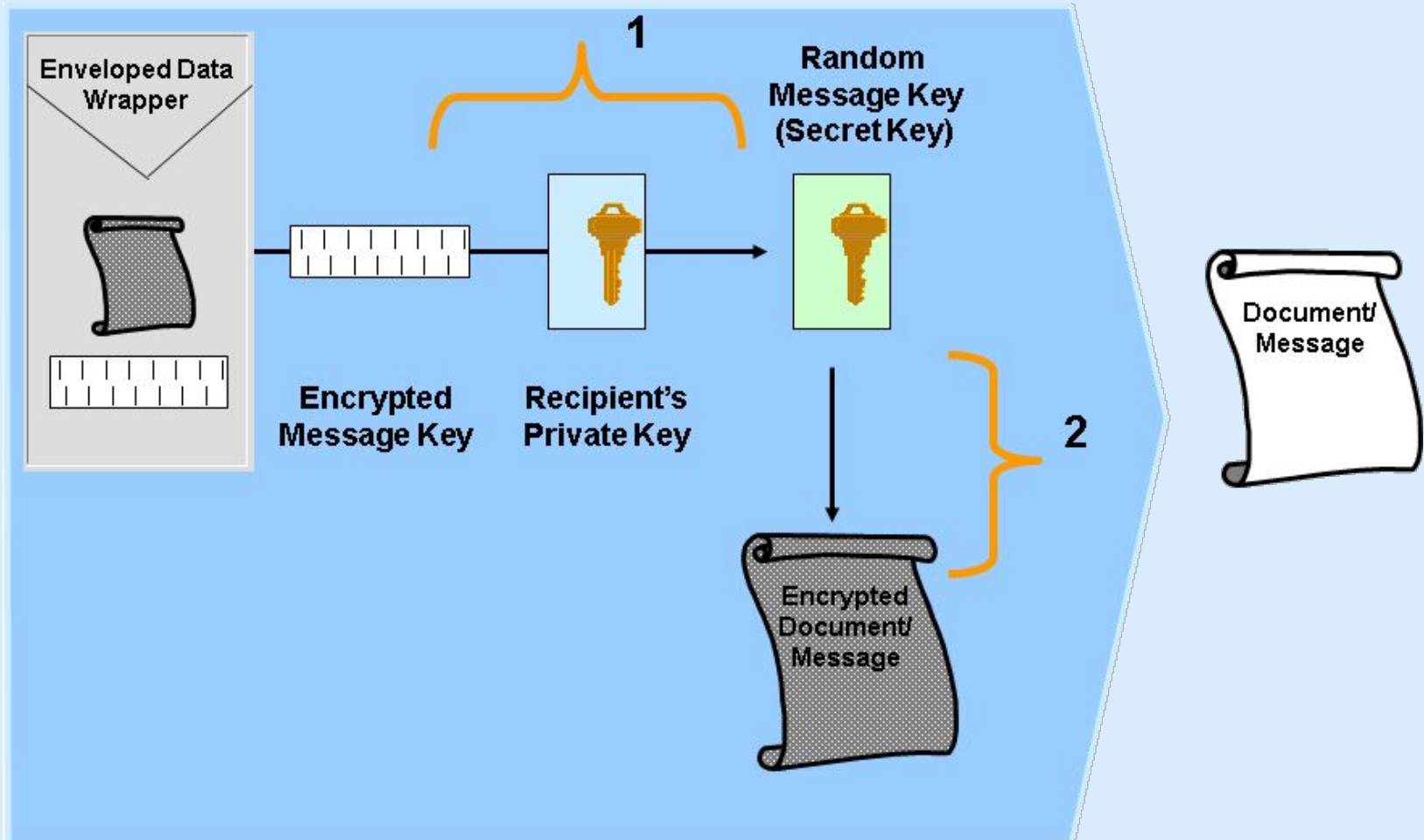
Digital Envelope - Creating



Digital Envelope - Opening



Received
→



Diffie-Hellman Key Exchange



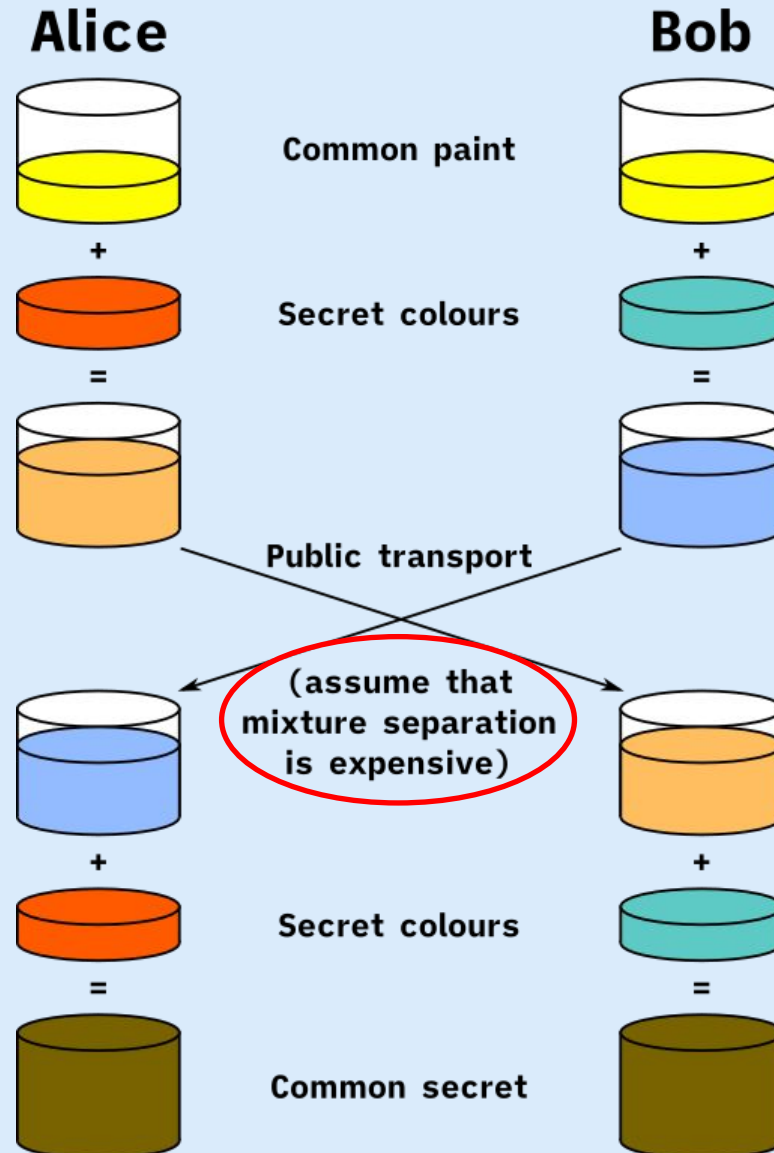
- First public key algorithm invented
 - Published in 1976
- Specific method for securely exchanging cryptographic keys over a public channel
- Named after inventors Whitfield Diffie and Martin Hellman
- It is an algorithm for **establishing shared secret key** — its NOT meant for encryption or signatures

Diffie-Hellman Algorithm

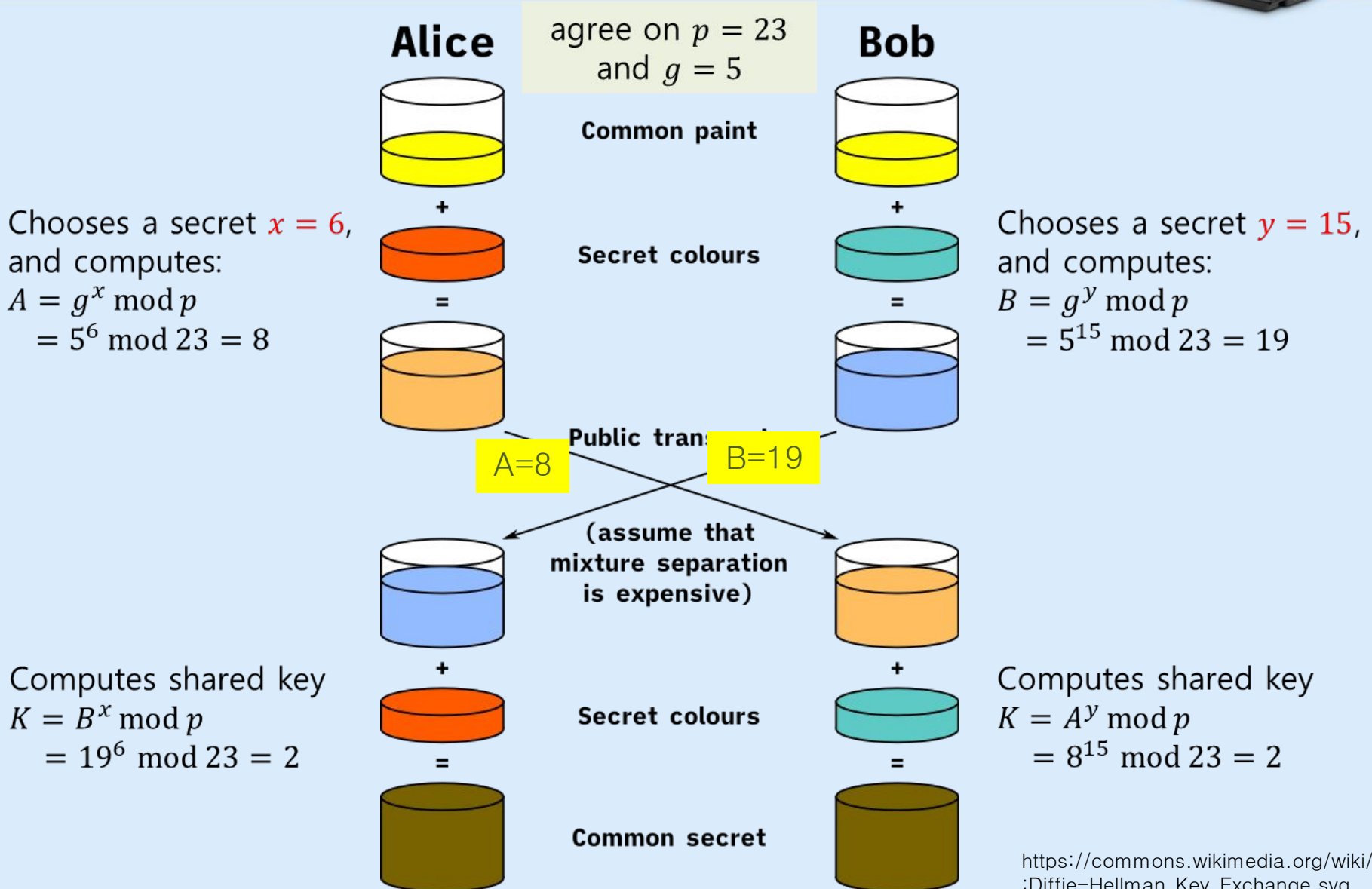


- Consider two numbers g and p shared publically between Alice and Bob.
 - p is a prime number
 - g , called generator, is a primitive root of p
 - $1 < g < p$
- Alice computes $A = g^x \bmod p$ (x is the secret with Alice)
- Bob computes $B = g^y \bmod p$ (y is the secret with Bob)
- Alice and Bob exchange A and B
- Alice computes $K_{Alice} = B^x \bmod p$
- Bob computes $K_{Bob} = A^y \bmod p$
- Now both have a shared secret, as $K_{Alice} = K_{Bob} = g^{xy} \bmod p$

Diffie-Hellman Paint Color Analogy



Diffie-Hellman Paint Color Analogy



Strength of Diffie-Hellman Algo



- An attacker watching all communication between Alice and Bob, will have access to their public numbers A and B
- Can they work out the secret numbers x & y ?
- It will require solving equations of the form $8 = 5^x \bmod 23$
- This is called **Discrete Logarithm Problem**
- When the prime p is large (≥ 2048 bits), solving this problem takes an infeasibly long time.

Inverse of exponentiation is log, inverse of 'modular exponentiation' is discrete log

Question



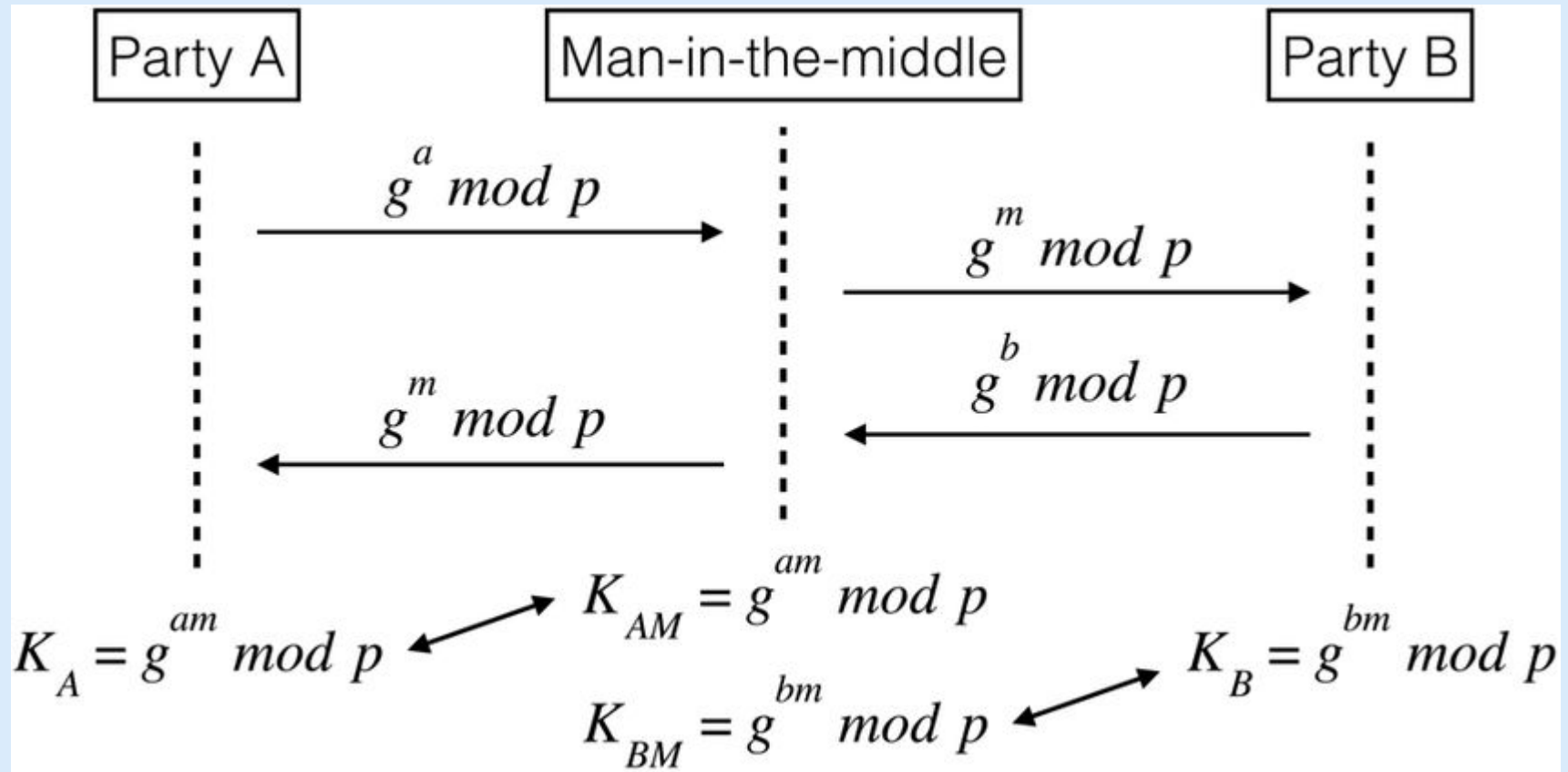
Both Diffie–Hellman algorithm and Digital Envelope allow us to combine the efficiency of symmetric encryption with the convenience of PKC (asymmetric).

- When would you prefer one over the other?

MITM against Diffie-Hellman



DH is vulnerable to man in the middle attack



MITM in PKC



- MITM is not unique to Diffie–Hellman key exchange
- All kinds of asymmetric crypto (RSA, digital signatures, digital envelope etc.) are vulnerable to such attacks
- Whenever public keys are exchanged over an insecure channel, we can not blindly trust the ‘authenticity’ of received public key
- Solution to this challenge is PKI
 - covered later

References



- Stallings CNS chap 9, 10
- RSA encryption more explanation with examples
<https://www.youtube.com/watch?v=UjIPMJd6Xks>
- RSA key generation online
https://www.mobilefish.com/services/rsa_key_generation/rsa_key_generation.php
- Diffie Hellman Key Exchange: youtu.be/85oMrKd8afY